Substituting $x_0 = ut$ in Eq. (6.28) and then evaluating the negative of the derivative of the flux with respect to time gives

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = -\frac{d}{dt}\left(\frac{B_0 l u^2 t^2}{2}\right) = -B_0 u^2 l t \quad \text{(V)}, \quad (6.29)$$

which is identical with Eq. (6.27). Since V_{12} is negative, the current $I = B_0 u^2 l t / R$ flows in the direction shown in **Fig. 6-8**.

Example 6-4: Moving Loop

The rectangular loop shown in **Fig. 6-9** is situated in the x–y plane and moves away from the origin with velocity $\mathbf{u} = \hat{\mathbf{y}}5$ (m/s) in a magnetic field given by

$$\mathbf{B}(y) = \hat{\mathbf{z}}0.2e^{-0.1y} \quad \text{(T)}.$$

If $R = 5\ \Omega$, find the current I at the instant that the loop sides are at $y_1 = 2$ m and $y_2 = 2.5$ m. The loop resistance may be ignored.

Solution: Since $\mathbf{u} \times \mathbf{B}$ is along $\hat{\mathbf{x}}$, voltages are induced across only the sides oriented along $\hat{\mathbf{x}}$, namely the sides linking point 1 to 2 and point 3 to 4. Had $\mathbf{B}$ been uniform, the induced voltages would have been the same, and the net voltage across the resistor would have been zero. In the present case, however, $\mathbf{B}$ decreases exponentially with y, thereby assuming a different value over side 1–2 than over side 3–4. Side 1–2 is at $y_1 = 2$ m, and the corresponding magnetic field is

$$\mathbf{B}(y_1) = \hat{\mathbf{z}}0.2e^{-0.1y_1} = \hat{\mathbf{z}}0.2e^{-0.2} \quad \text{(T)}.$$

The induced voltage V_{12} is then given by

$$\begin{aligned} V_{12} &= \int_2^1 [\mathbf{u} \times \mathbf{B}(y_1)] \cdot d\mathbf{l} \\ &= \int_{l/2}^{-l/2} (\hat{\mathbf{y}}5 \times \hat{\mathbf{z}}0.2e^{-0.2}) \cdot \hat{\mathbf{x}}\, dx \\ &= -e^{-0.2} l \\ &= -2e^{-0.2} \\ &= -1.637 \quad \text{(V)}. \end{aligned}$$

Similarly,

$$\begin{aligned} V_{43} &= -uB(y_2)\, l \\ &= -5 \times 0.2e^{-0.25} \times 2 \\ &= -1.558 \quad \text{(V)}. \end{aligned}$$

Consequently, the current is in the direction shown in the figure and its magnitude is

$$I = \frac{V_{43} - V_{12}}{R} = \frac{0.079}{5} = 15.8 \text{ (mA)}.$$

Example 6-5: Moving Rod Next to a Wire

The wire shown in **Fig. 6-10** carries a current $I = 10$ A. A 30-cm long metal rod moves with a constant velocity $\mathbf{u} = \hat{\mathbf{z}}5$ m/s. Find V_{12}, where 1 and 2 are the two ends of the rod.

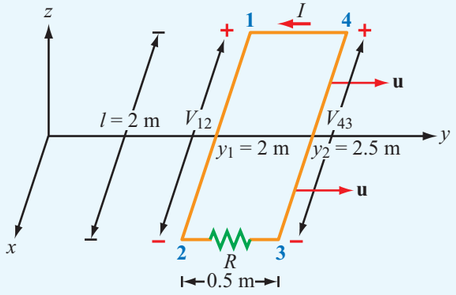


Figure 6-9 Moving loop of Example 6-4.

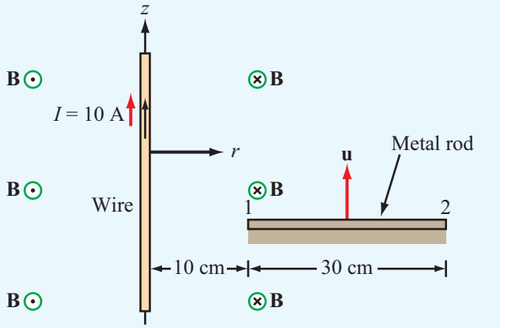


Figure 6-10 Moving rod of Example 6-5.

将 $x_0 = ut$ 代入等式：(6.28) 然后计算通量对时间的导数的负数给出

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = -\frac{d}{dt}\left(\frac{B_0 l u^2 t^2}{2}\right) = -B_0 u^2 l t \quad \text{(V)}, \quad (6.29)$$

这与等式相同。（6.27）。由于V_{12}为负，电流$I=B_0u^2lt/R$沿图6-8所示的方向流动。

例 6-4：移动循环

图 6-9 所示的矩形环位于 x-y 平面上，并在磁场中以速度 $\mathbf{u} = \hat{\mathbf{y}}5$ (m/s) 远离原点，该速度为：

$$\mathbf{B}(y) = \hat{\mathbf{z}}0.2e^{-0.1y} \quad \text{(T)}.$$

若$R = 5\ \Omega$，求环路边在$y_1 = 2$ m和$y_2 = 2.5$ m处瞬间的电流I。环路电阻可以忽略不计。

解：由于 u× B 沿 $\hat{\mathbf{x}}$，因此仅在沿 $\hat{\mathbf{x}}$ 定向的边（即连接点 1 到 2 和点 3 到 4 的边）上感应电压。如果 B 均匀，则感应电压将相同，电阻器两端的净电压将为零。然而，在本例中，B 随 y 呈指数下降，因此在 1-2 边上假设的值与在 3-4 边上的值不同。边 1–2 位于 $y_1 = 2$ m 处，对应的磁场为

$$\mathbf{B}(y_1) = \hat{\mathbf{z}}0.2e^{-0.1y_1} = \hat{\mathbf{z}}0.2e^{-0.2} \quad \text{(T)}.$$

感应电压 V_{12} 由下式给出

$$\begin{aligned} V_{12} &= \int_2^1 [\mathbf{u} \times \mathbf{B}(y_1)] \cdot d\mathbf{l} \\ &= \int_{l/2}^{-l/2} (\hat{\mathbf{y}}5 \times \hat{\mathbf{z}}0.2e^{-0.2}) \cdot \hat{\mathbf{x}}\, dx \\ &= -e^{-0.2} l \\ &= -2e^{-0.2} \\ &= -1.637 \quad \text{(V)}. \end{aligned}$$

相似地，

$$\begin{aligned} V_{43} &= -uB(y_2)\, l \\ &= -5 \times 0.2e^{-0.25} \times 2 \\ &= -1.558 \quad \text{(V)}. \end{aligned}$$

因此，电流的方向如图所示，其大小为

$$I = \frac{V_{43} - V_{12}}{R} = \frac{0.079}{5} = 15.8 \text{ (mA)}.$$

例 6-5：移动电线旁边的杆

图6-10所示的导线承载电流$I = 10$ A。一根30厘米长的金属棒以恒定速度$\mathbf{u} = \hat{\mathbf{z}}5$ m/s移动。求 V_{12}，其中 1 和 2 是杆的两端。

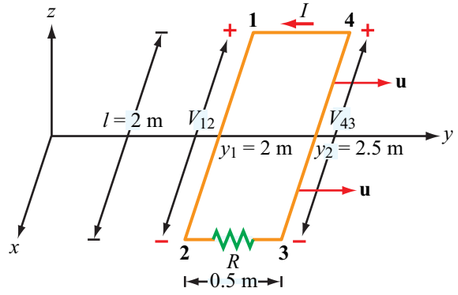


图 6-9 例 6-4 的移动循环。

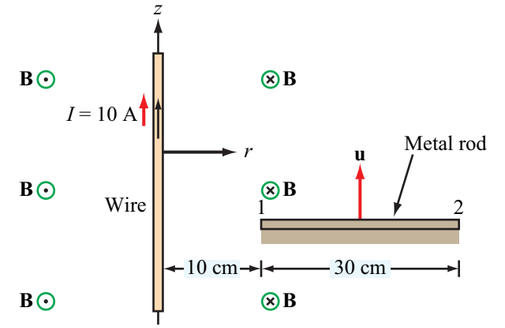
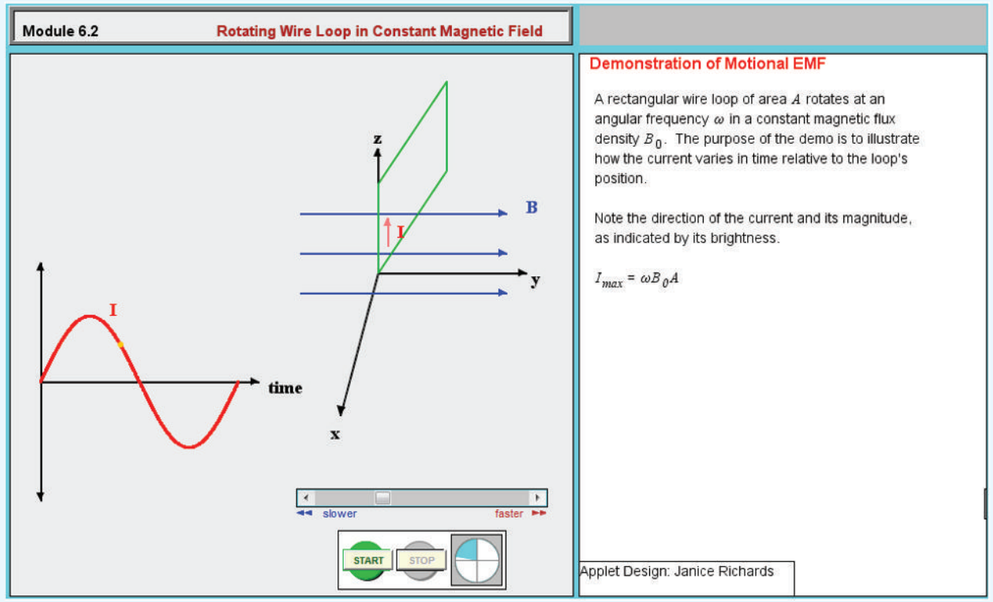


图6-10 例6-5的移动杆。

Module 6.2 Rotating Wire Loop in Constant Magnetic Field The principle of the electromagnetic generator is demonstrated by a rectangular loop rotating in the presence of a magnetic field.



Solution: The current I induces a magnetic field

$$\mathbf{B} = \hat{\boldsymbol{\phi}} \frac{\mu_0 I}{2\pi r},$$

where r is the radial distance from the wire and the direction of $\hat{\boldsymbol{\phi}}$ is into the page on the rod side of the wire. The movement of the rod in the presence of the field $\mathbf{B}$ induces a motional emf given by

$$\begin{aligned} V_{12} &= \int_{40\text{ cm}}^{10\text{ cm}} (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} \\ &= \int_{40\text{ cm}}^{10\text{ cm}} \left(\hat{\mathbf{z}} 5 \times \hat{\boldsymbol{\phi}} \frac{\mu_0 I}{2\pi r}\right) \cdot \hat{\mathbf{r}}\, dr \\ &= -\frac{5\mu_0 I}{2\pi} \int_{40\text{ cm}}^{10\text{ cm}} \frac{dr}{r} \\ &= -\frac{5 \times 4\pi \times 10^{-7} \times 10}{2\pi} \times \ln\left(\frac{10}{40}\right) \\ &= 13.9 \qquad (\mu\text{V}). \end{aligned}$$

Concept Question 6-4: Suppose that no friction is involved in sliding the conducting bar of **Fig. 6-8** and that the horizontal arms of the circuit are very long. Hence, if the bar is given an initial push, it should continue moving at a constant velocity, and its movement generates electrical energy in the form of an induced emf, indefinitely. Is this a valid argument? If not, why not? Can we generate electrical energy without having to supply an equal amount of energy by other means?

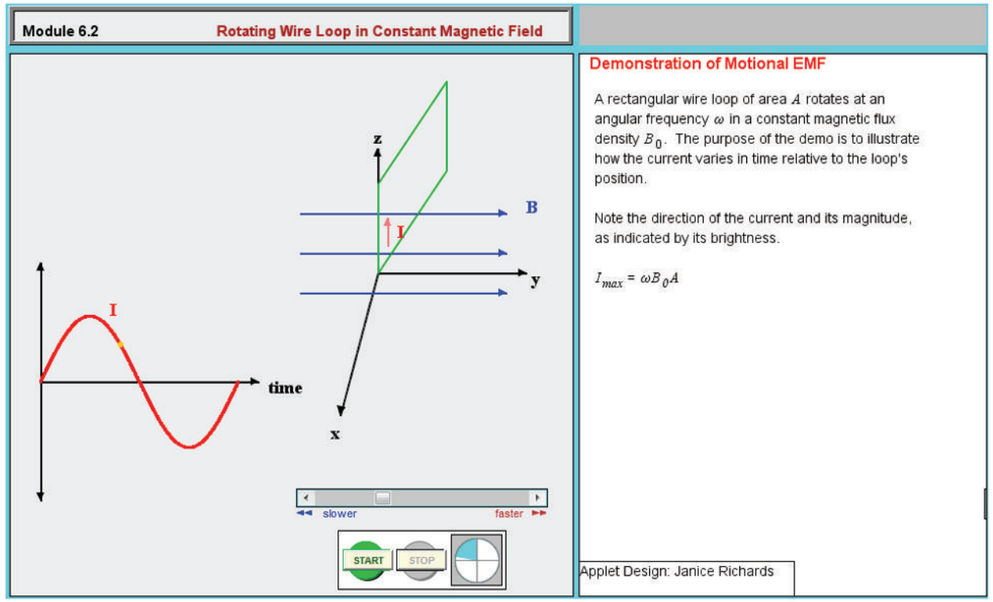
Concept Question 6-5: Is the current flowing in the rod of **Fig. 6-10** a steady current? Examine the force on a charge q at ends 1 and 2 and compare.

Exercise 6-3: For the moving loop of **Fig. 6-9**, find I when the loop sides are at $y_1 = 4$ m and $y_2 = 4.5$ m. Also, reverse the direction of motion such that $\mathbf{u} = -\hat{\mathbf{y}}5$ (m/s).

Answer: $I = -13$ (mA). (See EM.)

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模块 6.2 恒定磁场中的旋转线圈 电磁发电机的原理通过在磁场中旋转的矩形线圈来演示。



解：电流I感应出磁场

$$\mathbf{B} = \hat{\boldsymbol{\phi}} \frac{\mu_0 I}{2\pi r},$$

其中 r 是距线材的径向距离，$\hat{\boldsymbol{\phi}}$ 的方向是线材杆侧进入页面的方向。在磁场 B 存在的情况下，杆的运动会产生由下式给出的运动电动势：

$$\begin{aligned} V_{12} &= \int_{40\text{ cm}}^{10\text{ cm}} (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} \\ &= \int_{40\text{ cm}}^{10\text{ cm}} \left(\hat{\mathbf{z}} 5 \times \hat{\boldsymbol{\phi}} \frac{\mu_0 I}{2\pi r}\right) \cdot \hat{\mathbf{r}}\, dr \\ &= -\frac{5\mu_0 I}{2\pi} \int_{40\text{ cm}}^{10\text{ cm}} \frac{dr}{r} \\ &= -\frac{5 \times 4\pi \times 10^{-7} \times 10}{2\pi} \times \ln\left(\frac{10}{40}\right) \\ &= 13.9 \qquad (\mu\text{V}). \end{aligned}$$

概念问题 6-4：假设图 6-8 中的导电棒滑动时不存在摩擦力，并且电路的水平臂很长。因此，如果杆受到初始推力，它应该继续以恒定速度移动，并且其运动无限期地以感应电动势的形式产生电能。这是一个有效的论点吗？如果没有，为什么不呢？我们能否在无需通过其他方式提供等量能源的情况下产生电能？

概念问题 6-5：图 6-10 棒中流动的电流是稳定电流吗？检查末端 1 和 2 处电荷 q 上的力并进行比较。

练习6-3：对于图6-9的移动环，求环边位于$y_1 = 4$ m和$y_2 = 4.5$ m处时的I。另外，反转运动方向，使得 u $= -\hat{\mathbf{y}}5$ (m/s)。

答案：我$= -13$（毫安）。（参见EM。）

Exercise 6-4: Suppose that we turn the loop of **Fig. 6-9** so that its surface is parallel to the x–z plane. What would I be in that case?

Answer: $I = 0$. (See EM.)

6-5 The Electromagnetic Generator

The electromagnetic generator is the converse of the electromagnetic motor. The principles of operation of both instruments may be explained with the help of **Fig. 6-11**. A permanent magnet is used to produce a static magnetic field **B** in the slot between its two poles. When a current is passed through the conducting loop, as depicted in **Fig. 6-11(a)**, the current flows in opposite directions in segments 1–2 and 3–4 of the loop. The induced magnetic forces on the two segments are also opposite, resulting in a torque that causes the loop to rotate about its axis. Thus, in a motor, electrical energy supplied by a voltage source is converted into mechanical energy in the form of a rotating loop, which can be coupled to pulleys, gears, or other movable objects.

If instead of passing a current through the loop to make it turn the loop is made to rotate by an external force, the movement of the loop in the magnetic field produces a motional emf, $V^{\mathrm{m}}_{\mathrm{emf}}$, as shown in **Fig. 6-11(b)**. Hence, the motor has become a generator, and mechanical energy is getting converted into electrical energy.

Let us examine the operation of the electromagnetic generator in more detail using the coordinate system shown in **Fig. 6-12**. The magnetic field is

$$\mathbf{B} = \hat{\mathbf{z}} B_0, \tag{6.30}$$

and the axis of rotation of the conducting loop is along the x axis. Segments 1–2 and 3–4 of the loop are of length l each, and both cross the magnetic flux lines as the loop rotates. The other two segments are each of width w, and neither crosses the **B** lines when the loop rotates. Hence, only segments 1–2 and 3–4 contribute to the generation of the motional emf, $V^{\mathrm{m}}_{\mathrm{emf}}$.

As the loop rotates with an angular velocity ω about its own axis, segment 1–2 moves with velocity **u** given by

$$\mathbf{u} = \hat{\mathbf{n}} \omega \frac{w}{2}, \tag{6.31}$$

where $\hat{\mathbf{n}}$, the surface normal to the loop, makes an angle α with the z axis. Hence,

$$\hat{\mathbf{n}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}} \sin\alpha. \tag{6.32}$$

Segment 3–4 moves with velocity $-\mathbf{u}$. Application of

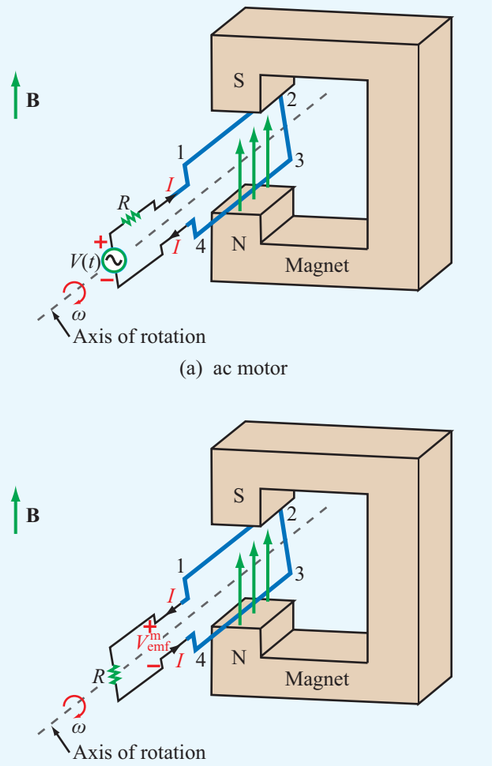


Figure 6-11 Principles of the ac motor and the ac generator. In (a) the magnetic torque on the wires causes the loop to rotate and in (b) the rotating loop generates an emf.

Eq. (6.26), consistent with our choice of $\hat{\mathbf{n}}$, gives

$$\begin{aligned} V^{\mathrm{m}}_{\mathrm{emf}} = V_{14} &= \int_2^1 (\mathbf{u}\times\mathbf{B})\cdot d\mathbf{l} + \int_4^3 (\mathbf{u}\times\mathbf{B})\cdot d\mathbf{l} \\ &= \int_{-l/2}^{l/2} \left[\left(\hat{\mathbf{n}}\omega\frac{w}{2}\right)\times\hat{\mathbf{z}}B_0\right]\cdot\hat{\mathbf{x}}\,dx \\ &\quad + \int_{l/2}^{-l/2} \left[\left(-\hat{\mathbf{n}}\omega\frac{w}{2}\right)\times\hat{\mathbf{z}}B_0\right]\cdot\hat{\mathbf{x}}\,dx. \end{aligned} \tag{6.33}$$

Using Eq. (6.32) in Eq. (6.33), we obtain the result

$$V^{\mathrm{m}}_{\mathrm{emf}} = wl\omega B_0 \sin\alpha = A\omega B_0 \sin\alpha, \tag{6.34}$$

练习 6-4：假设我们转动图 6-9 中的环，使其表面平行于 x-z 平面。那样的话我会怎样呢？

答案：我 $= 0$。（参见 EM 。）

6-5 电磁发生器

电磁发电机与电磁电机相反。两种仪器的工作原理可以借助图 6-11 进行解释。永磁体用于在其两极之间的槽中产生静磁场 B。当电流通过导电回路时，如图6-11(a)所示，电流在回路的1-2和3-4段中以相反方向流动。两个部分上的感应磁力也是相反的，从而产生使环绕其轴旋转的扭矩。因此，在电机中，由电压源提供的电能以旋转环的形式转换为机械能，该旋转环可以耦合到滑轮、齿轮或其他可移动物体。

如果不是让电流通过环路使其转动，而是通过外力使环路旋转，则环路在磁场中的运动会产生运动电动势 $V^{\mathrm{m}}_{\mathrm{emf}}$，如图 6-11(b) 所示。因此，电动机变成了发电机，机械能转化为电能。

让我们使用图 6-12 所示的坐标系更详细地检查电磁发电机的操作。磁场是

$$\mathbf{B} = \hat{\mathbf{z}} B_0, \tag{6.30}$$

导电环的旋转轴沿x轴。环路的 1-2 段和 3-4 段的长度各为 l，并且在环路旋转时都穿过磁力线。另外两段的宽度均为 w，当环旋转时，都不与 B 线相交。因此，只有段 1-2 和 3-4 有助于生成运动电动势 $V^{\mathrm{m}}_{\mathrm{emf}}$。

当环以角速度 ω 绕其自身轴旋转时，段 1-2 以速度 u 移动，速度 u 为：

$$\mathbf{u} = \hat{\mathbf{n}} \omega \frac{w}{2}, \tag{6.31}$$

其中，n̂（与环垂直的表面）与 z 轴形成角度 α。因此，

$$\hat{\mathbf{n}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}} \sin\alpha. \tag{6.32}$$

段 3-4 以速度 −u 移动。应用

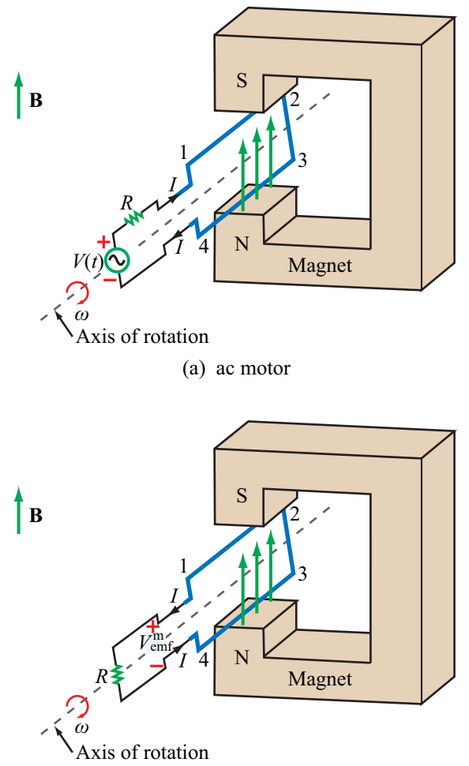


图6-11 交流电动机和交流发电机的原理。在（a）中，电线上的磁扭矩导致环路旋转，在（b）中，旋转环路产生电动势。

等式。(6.26) 与我们对 n̂ 的选择一致，给出

$$\begin{aligned} V^{\mathrm{m}}_{\mathrm{emf}} = V_{14} &= \int_2^1 (\mathbf{u}\times\mathbf{B})\cdot d\mathbf{l} + \int_4^3 (\mathbf{u}\times\mathbf{B})\cdot d\mathbf{l} \\ &= \int_{-l/2}^{l/2} \left[\left(\hat{\mathbf{n}}\omega\frac{w}{2}\right)\times\hat{\mathbf{z}}B_0\right]\cdot\hat{\mathbf{x}}\,dx \\ &\quad + \int_{l/2}^{-l/2} \left[\left(-\hat{\mathbf{n}}\omega\frac{w}{2}\right)\times\hat{\mathbf{z}}B_0\right]\cdot\hat{\mathbf{x}}\,dx. \end{aligned} \tag{6.33}$$

使用方程式(6.32) 式（6.33），我们得到结果

$$V^{\mathrm{m}}_{\mathrm{emf}} = wl\omega B_0 \sin\alpha = A\omega B_0 \sin\alpha, \tag{6.34}$$

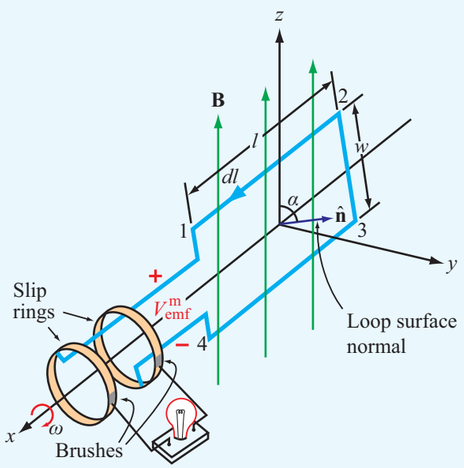


Figure 6-12 A loop rotating in a magnetic field induces an emf.

where $A = wl$ is the surface area of the loop. The angle α is related to ω by

$$\alpha = \omega t + C_0, \tag{6.35}$$

where C_0 is a constant determined by initial conditions. For example, if $\alpha = 0$ at $t = 0$, then $C_0 = 0$. In general,

$$V_{emf}^{m} = A\omega B_0 \sin(\omega t + C_0) \quad \text{(V)}. \tag{6.36}$$

This same result can also be obtained by applying the general form of Faraday's law given by Eq. (6.6). The flux linking the surface of the loop is

$$\Phi = \int_S \mathbf{B}\cdot d\mathbf{s} = \int_S \hat{\mathbf{z}}B_0\cdot\hat{\mathbf{n}}\,ds = B_0 A\cos\alpha = B_0 A\cos(\omega t + C_0), \tag{6.37}$$

and

$$V_{emf} = -\frac{d\Phi}{dt} = -\frac{d}{dt}[B_0 A\cos(\omega t + C_0)] = A\omega B_0\sin(\omega t + C_0), \tag{6.38}$$

which is identical with the result given by Eq. (6.36).

► The voltage induced by the rotating loop is sinusoidal in time with an angular frequency ω equal to that of the rotating loop, and its amplitude is equal to the product of the surface area of the loop, the magnitude of the magnetic field generated by the magnet, and the angular frequency ω. ◄

Concept Question 6-6: Contrast the operation of an ac motor with that of an ac generator.

Concept Question 6-7: The rotating loop of **Fig. 6-12** had a single turn. What would be the emf generated by a loop with 10 turns?

Concept Question 6-8: The magnetic flux linking the loop shown in **Fig. 6-12** is maximum when $\alpha = 0$ (loop in x–y plane), yet according to Eq. (6.34), the induced emf is zero when $\alpha = 0$. Conversely, when $\alpha = 90°$, the flux linking the loop is zero, but V_{emf}^{m} is at a maximum. Is this consistent with your expectations? Why?

6-6 Moving Conductor in a Time-Varying Magnetic Field

For the general case of a single-turn conducting loop moving in a time-varying magnetic field, the induced emf is the sum of a transformer component and a motional component. Thus, the sum of Eqs. (6.8) and (6.26) gives

$$V_{emf} = V_{emf}^{tr} + V_{emf}^{m} = -\int_S \frac{\partial \mathbf{B}}{\partial t}\cdot d\mathbf{s} + \oint_C (\mathbf{u}\times\mathbf{B})\cdot d\mathbf{l}. \tag{6.39}$$

V_{emf} is also given by the general expression of Faraday's law:

$$V_{emf} = -\frac{d\Phi}{dt} = -\frac{d}{dt}\int_S \mathbf{B}\cdot d\mathbf{s} \quad \text{(total emf)}. \tag{6.40}$$

In fact, it can be shown mathematically that the right-hand side of Eq. (6.39) is equivalent to the right-hand side of Eq. (6.40). For a particular problem, the choice between using Eq. (6.39) or Eq. (6.40) is usually made on the basis of which is the easier to apply. In either case, for an N-turn loop, the right-hand sides of Eqs. (6.39) and (6.40) should be multiplied by N.

Example 6-6: Electromagnetic Generator

Find the induced voltage when the rotating loop of the electromagnetic generator of Section 6-5 is in a magnetic field $\mathbf{B} = \hat{\mathbf{z}}B_0\cos\omega t$. Assume that $\alpha = 0$ at $t = 0$.

Solution: The flux Φ is given by Eq. (6.37) with B_0 replaced with $B_0\cos\omega t$. Thus,

$$\Phi = B_0 A\cos^2\omega t,$$

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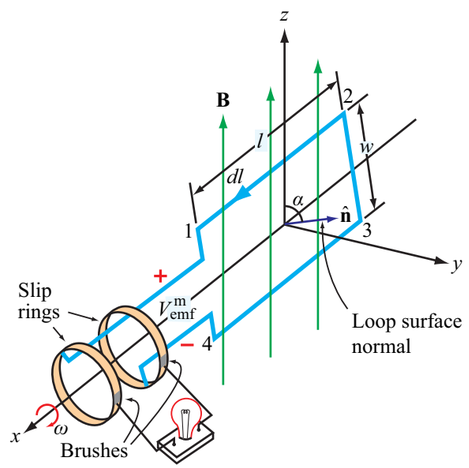


图 6-12 在磁场中旋转的环路会产生电动势。

概念问题 6-6：比较交流电动机和交流发电机的运行。

概念题 6-7：图 6-12 的旋转环只有一圈。 10 匝环路产生的电动势是多少？

概念问题 6-8：当 α = 0（x-y 平面中的环路）时，连接图 6-12 所示环路的磁通量最大，但根据方程：根据式(6.34)，当α =为0时，感应电动势为零。相反，当α =为90°时，连接环路的磁通为零，但V_{emf}^{m}达到最大值。这符合您的预期吗？为什么？

6-6 时变磁场中移动的导体

对于在时变磁场中移动的单匝导电环路的一般情况，感应电动势是变压器分量和运动分量的总和。因此，等式的总和。 (6.8) 和 (6.26) 给出

$$V_{emf} = V_{emf}^{tr} + V_{emf}^{m} = -\int_S \frac{\partial \mathbf{B}}{\partial t}\cdot d\mathbf{s} + \oint_C (\mathbf{u}\times\mathbf{B})\cdot d\mathbf{l}. \tag{6.39}$$

Vemf 也由法拉第定律的一般表达式给出：

$$V_{emf} = -\frac{d\Phi}{dt} = -\frac{d}{dt}\int_S \mathbf{B}\cdot d\mathbf{s} \quad \text{(total emf)}. \tag{6.40}$$

事实上，可以从数学上证明方程的右侧： (6.39) 等价于式 (6.39) 的右侧。 （6.40）。对于特定问题，可以选择使用方程。 （6.39）或等式。 (6.40)通常是在更容易应用的基础上制定的。在任何一种情况下，对于 N 匝循环，方程的右侧。 (6.39)和(6.40)应乘以N。

其中 A = wl 是环路的表面积。角度 α 与 ω 的关系为

$$\alpha = \omega t + C_0, \tag{6.35}$$

其中 C_0 是由初始条件确定的常数。例如，如果在 t = 0 处 α = 0，则 $C_0 = 0$。一般来说，

$$V_{emf}^{m} = A\omega B_0\sin(\omega t + C_0) \quad \text{(V)}. \tag{6.36}$$

通过应用等式给出的法拉第定律的一般形式也可以获得同样的结果。 （6.6）。连接环路表面的通量为

$$\Phi = \int_S \mathbf{B}\cdot d\mathbf{s} = \int_S \hat{\mathbf{z}}B_0\cdot\hat{\mathbf{n}}\,ds = B_0 A\cos\alpha = B_0 A\cos(\omega t + C_0), \tag{6.37}$$

和

$$V_{emf} = -\frac{d\Phi}{dt} = -\frac{d}{dt}[B_0 A\cos(\omega t + C_0)] = A\omega B_0\sin(\omega t + C_0), \tag{6.38}$$

这与等式给出的结果相同。 （6.36）。

► 旋转环感应的电压随时间呈正弦波，其角频率ω等于旋转环的电压，其振幅等于环的表面积、磁体产生的磁场强度和角频率ω的乘积。 ◄

例 6-6：电磁发生器

求第 6-5 节的电磁发电机的旋转回路处于磁场 $\mathbf{B} = \hat{\mathbf{z}}B_0\cos\omega t$ 中时的感应电压。假设 t = 0 时 α = 0。

解： 通量 Φ 由方程式给出。 (6.37) 将 B_0 替换为 $B_0\cos\omega t$。因此，

$$\Phi = B_0 A\cos^2\omega t,$$

and

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = -\frac{d}{dt}(B_0 A\cos^2\omega t) = 2B_0 A\omega\cos\omega t\sin\omega t = B_0 A\omega\sin 2\omega t.$$

6-7 Displacement Current

Ampère's law in differential form is given by

$$\nabla\times\mathbf{H} = \mathbf{J} + \frac{\partial\mathbf{D}}{\partial t} \qquad \textbf{(Ampère's law)}. \tag{6.41}$$

Integrating both sides of Eq. (6.41) over an arbitrary open surface S with contour C, we have

$$\int_S (\nabla\times\mathbf{H})\cdot d\mathbf{s} = \int_S \mathbf{J}\cdot d\mathbf{s} + \int_S \frac{\partial\mathbf{D}}{\partial t}\cdot d\mathbf{s}. \tag{6.42}$$

The surface integral of $\mathbf{J}$ equals the conduction current I_c flowing through S, and the surface integral of $\nabla\times\mathbf{H}$ can be converted into a line integral of $\mathbf{H}$ over the contour C bounding C by invoking Stokes's theorem. That is,

$$\int_S \mathbf{J}\cdot d\mathbf{s} = I_c,$$

$$\int_S (\nabla\times\mathbf{H})\cdot d\mathbf{s} = \oint_C \mathbf{H}\cdot d\mathbf{l},$$

and

$$\oint_C \mathbf{H}\cdot d\mathbf{l} = I_c + \int_S \frac{\partial\mathbf{D}}{\partial t}\cdot d\mathbf{s}. \tag{6.43}$$

(Ampère's law)

The second term on the right-hand side of Eq. (6.43) of course has the same unit (amperes) as the current I_c, and because it is proportional to the time derivative of the electric flux density $\mathbf{D}$, which is also called the electric displacement, it is called the ***displacement current*** I_d. That is,

$$I_d = \int_S \mathbf{J}_d\cdot d\mathbf{s} = \int_S \frac{\partial\mathbf{D}}{\partial t}\cdot d\mathbf{s}, \tag{6.44}$$

where $\mathbf{J}_d = \partial\mathbf{D}/\partial t$ represents a ***displacement current density***. In view of Eq. (6.44),

$$\oint_C \mathbf{H}\cdot d\mathbf{l} = I_c + I_d = I, \tag{6.45}$$

where I is the total current. In electrostatics, $\partial\mathbf{D}/\partial t = 0$; therefore, $I_d = 0$ and $I = I_c$. The concept of displacement current was first introduced in 1873 by James Clerk Maxwell when he formulated the unified theory of electricity and magnetism under time-varying conditions.

The parallel-plate capacitor is commonly used as an example with which to illustrate the physical meaning of the displacement current I_d. The simple circuit shown in **Fig. 6-13** consists of a capacitor and an ac source with voltage $V_s(t)$ given by

$$V_s(t) = V_0\cos\omega t \qquad \text{(V)}. \tag{6.46}$$

According to Eq. (6.45), the total current flowing through any surface consists, in general, of a conduction current I_c and a displacement current I_d. Let us find I_c and I_d through each of the following two imaginary surfaces: (1) the cross section of the conducting wire, S_1, and (2) the cross section of the capacitor S_2 (**Fig. 6-13**). We denote the conduction and displacement currents in the wire as I_{1c} and I_{1d} and those through the capacitor as I_{2c} and I_{2d}.

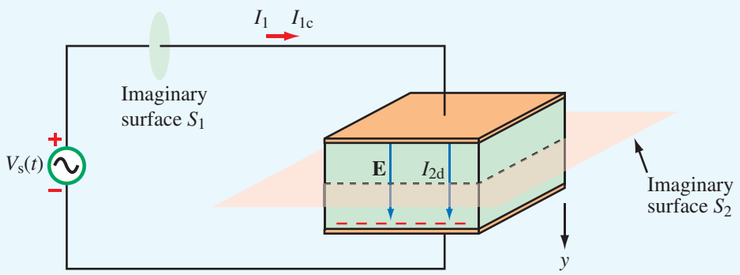


Figure 6-13 The displacement current I_{2d} in the insulating material of the capacitor is equal to the conducting current I_{1c} in the wire.

和

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = -\frac{d}{dt}(B_0 A\cos^2\omega t) = 2B_0 A\omega\cos\omega t\sin\omega t = B_0 A\omega\sin 2\omega t.$$

6-7 位移电流

微分形式的安培定律　　由下式给出

$$\nabla\times\mathbf{H} = \mathbf{J} + \frac{\partial\mathbf{D}}{\partial t} \qquad \textbf{(Ampère's law)}. \tag{6.41}$$

对方程两边求积分。 (6.41) 在具有轮廓 C 的任意开放曲面 S 上，我们有

$$\int_S (\nabla\times\mathbf{H})\cdot d\mathbf{s} = \int_S \mathbf{J}\cdot d\mathbf{s} + \int_S \frac{\partial\mathbf{D}}{\partial t}\cdot d\mathbf{s}. \tag{6.42}$$

J 的表面积分等于流过 S 的传导电流 I_c，并且 $\nabla\times$ H 的表面积分可以通过调用斯托克斯定理转换为 H 在以 C 为界的轮廓 C 上的线积分。那是，

$$\int_S \mathbf{J}\cdot d\mathbf{s} = I_c,$$

$$\int_S (\nabla\times\mathbf{H})\cdot d\mathbf{s} = \oint_C \mathbf{H}\cdot d\mathbf{l},$$

和

$$\oint_C \mathbf{H}\cdot d\mathbf{l} = I_c + \int_S \frac{\partial\mathbf{D}}{\partial t}\cdot d\mathbf{s}. \tag{6.43}$$

(安培定律　　)

等式右侧的第二项。式（6.43）当然与电流I_c具有相同的单位（安培），并且由于它与电通量密度D的时间导数（也称为电位移）成正比，因此称为位移电流I_d。那是，

$$I_d = \int_S \mathbf{J}_d\cdot d\mathbf{s} = \int_S \frac{\partial\mathbf{D}}{\partial t}\cdot d\mathbf{s}, \tag{6.44}$$

其中，$J_d = \partial D/\partial t$表示位移电流密度。鉴于方程。 (6.44),

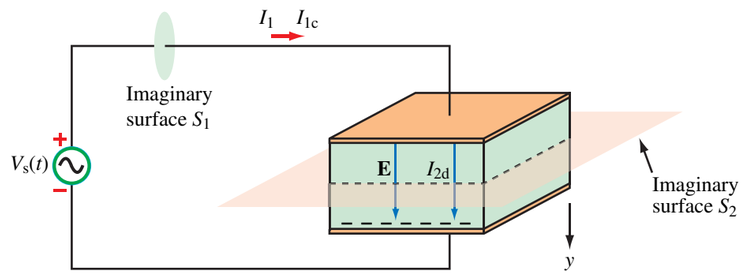
$$\oint_C \mathbf{H}\cdot d\mathbf{l} = I_c + I_d = I, \tag{6.45}$$

其中 I 是总电流。在静电学中，$\partial D/\partial t = 0$；因此，$I_d = 0$ 和 $I = I_c$。位移电流的概念由詹姆斯·克拉克·麦克斯韦 (James Clerk Maxwell) 于 1873 年首次提出，当时他制定了时变条件下的电和磁的统一理论。

通常以平行板电容器为例来说明位移电流I_d的物理意义。图 6-13 所示的简单电路由一个电容器和一个交流电源组成，电压 $V_s(t)$ 为

$$V_s(t) = V_0\cos\omega t \qquad \text{(V)}. \tag{6.46}$$

根据方程。根据方程（6.45），流过任何表面的总电流通常由传导电流 I_c 和位移电流 I_d 组成。让我们通过以下两个假想表面分别求出 I_c 和 I_d：(1) 导线的横截面 S_1，以及 (2) 电容器的横截面 S_2（。 6-13)。我们将导线中的传导电流和位移电流表示为 I_{1c} 和 I_{1d}，将通过电容器的传导电流和位移电流表示为 I_{2c} 和 I_{2d}。



F图 6-13 电容器绝缘材料中的位移电流 I_{2d} 等于导线中的传导电流 I_{1c}。